# Literature Review

Real-Time Chemotherapy Patient Monitoring with AI-Based Anomaly Detection

## 1. Context and Review Purpose

Monitoring chemotherapy patients requires clear symptom reporting, timely access to observations, and reliable processing of information from different sources. This focused review synthesizes three selected studies: a chemotherapy monitoring trial [1], a synthetic physiological streaming evaluation [2], and a streaming-plus-deep-learning framework [3]. It establishes context, compares methodological limits, justifies design choices and identifies a narrowly stated project contribution. It is not a systematic review or an exhaustive literature search.

## 2. Critical Review of Selected Studies

### 2.1. Maguire et al. (2021): Clinical Monitoring and Alerts

The eSMART randomised trial evaluated the mobile Advanced Symptom Management System (ASyMS) with 829 patients in five European countries [1]. Patients reported chemotherapy symptoms and temperature remotely, and evidence-based clinical rules generated alerts for staff. The intervention was associated with lower symptom burden (adjusted MSAS difference -0.15) and improved health-related quality of life versus usual care. However, the study did not demonstrate reduced hospital admissions or deaths, and the monitoring setting was primarily remote rather than an in-hospital streaming pipeline. The evaluated system did not report a trained ML alert model. For our prototype, the useful lesson is an auditable patient-to-alert-to-review workflow, not a claim that ASyMS technology or clinical outcomes are reproduced.

### 2.2. Zablah et al. (2026): Physiological Streaming Architecture

Zablah et al. [2] presented an IoT biomedical architecture using simulated heart rate, oxygen saturation and temperature; MQTT/Mosquitto for communication; Kafka for ingestion; and Spark Structured Streaming for processing. Their anomaly logic used a 60-second moving average and predefined thresholds. The authors reported 47 seconds mean detection delay for streaming versus 300 seconds for periodic sampling in controlled simulation. Although the introduction mentions machine learning, the implemented detection described in the methods is threshold-based; learned anomaly models are proposed for future work. The paper supplies an architecture and a latency-evaluation example, but it does not establish clinical performance for real chemotherapy patients. Its reported communication tail-latency figures also differ between the abstract and the results, so the numerical claims require care.

### 2.3. Mallesh et al. (2026): ML within a Streaming Pipeline

Mallesh et al. [3] described a six-layer remote-monitoring framework integrating Kafka, Spark and a hierarchical LSTM trained to classify eight physiological states. The authors reported 1,440 samples from 20 participants, an 80/20 train/validation split, 98.70% validation accuracy and a mean 1.2-second alert delay. This is the selected paper that describes a trained deep-learning classifier, unlike [1] and [2]. Nevertheless, the experiment is small, single-site and not specific to chemotherapy. The published confusion-matrix graphic includes negative entries, which cannot represent ordinary count values; this reporting issue warrants caution before treating its accuracy as independently verified. The paper demonstrates a possible streaming-ML integration pattern, not proof that a chemotherapy monitoring model will have the same performance.

## 3. Comparative Synthesis

| Study | Research focus | AI implemented? | Evidence and limitation |
|---|---|---|---|
| [1] Maguire 2021 | Patient-reported chemotherapy symptoms; clinical review | No trained ML model reported; fixed clinical rules | Randomised clinical trial; predominantly remote follow-up. |
| [2] Zablah 2026 | IoT vital-sign transport and stream-processing speed | No; moving averages and thresholds | Synthetic load/performance measurements; not clinically validated. |
| [3] Mallesh 2026 | Multi-modal monitoring with Kafka, Spark and LSTM | Yes; hierarchical LSTM classifier | Small, single-site dataset; anomalous figure values reduce confidence. |

Cross-study interpretation: the clinical trial [1] motivates usable symptom review; the streaming study [2] informs infrastructure and event-time evaluation; the ML study [3] provides an example of learned detection in a pipeline. They evaluate different populations, input types and outcomes, so speed and accuracy figures cannot be compared as equivalent results or promised for this project.

## 4. Research Gap and Justification

Within these three selected works, none directly evaluates the exact educational combination proposed here: in-hospital chemotherapy sessions linked to synthetic device observations, nursing symptom entries, laboratory timestamps, a versioned Isolation Forest anomaly signal and append-only clinician-review history. This is a gap in the scope of the reviewed selection, not proof that no similar system exists elsewhere.

The proposed design adapts event streaming ideas [2,3] to a chemotherapy-focused workflow motivated by [1]. Unlike reproducing the LSTM, the team will select a simpler Isolation Forest model, trained and calibrated offline, as a feasible anomaly-detection experiment on engineered vital-sign windows; that choice is a new project decision, not a method clinically validated by the selected papers. Independent rule-based alerts will remain active. ANC and selected CBC results will be displayed as contextual test information, not inferred from vital signs or fed to the initial anomaly model.

## 5. Conceptual Framework and Evaluation

Conceptual data flow: synthetic vital-sign, laboratory and nursing events -> Kafka -> Spark data validation and event-time handling -> PostgreSQL -> five-minute vital-sign feature windows -> separate Python worker applying a saved, offline-trained Isolation Forest -> traceable dashboard flag and human review. The model starts from normal-pattern synthetic training data; calibration defines its cutoff, while patient-disjoint held-out scenarios test its response. Evaluation will report message integrity, lateness, recovery, processing time, scoring coverage, precision/recall, false anomaly episodes and detection delay.

This approach reduces duplication of earlier work: the prototype does not rebuild ASyMS, and it does not claim the same clinical outcomes or replicate the reported LSTM accuracy. Its contribution is a reproducible integration and testing exercise tailored to in-hospital chemotherapy monitoring. All simulated anomaly flags indicate unusual patterns for review, not verified clinical diagnoses or automatic treatment decisions.

## 6. References

[1] R. Maguire et al., “Real time remote symptom monitoring during chemotherapy for cancer: European multicentre randomised controlled trial (eSMART),” BMJ, 374:n1647, 2021. https://doi.org/10.1136/bmj.n1647

[2] I. Zablah et al., “IoT-Based Biomedical Architecture with Real-Time Streaming Analytics for Preventive Health Monitoring,” Proc. 24th LACCEI International Multi-Conference, 2026. https://doi.org/10.18687/LACCEI2026.1.1.1226

[3] S. Mallesh et al., “Big Data Analytics for Remote Patient Monitoring in Cyber-Physical Healthcare Systems,” Engineering, Technology & Applied Science Research, 16(4), 37571-37576, 2026. https://doi.org/10.48084/etasr.18957